S2S-01: Step 1 — Discover: Migration Scenarios and Inventory

> **Series:** S2S | **Notebook:** 1 of 10 | **Phase:** Plan | **Step:** Discover | **Created:** March 2026 | **Last Updated:** 04/16/2026

The first step in any SaaS-to-SaaS migration is understanding *why* you are migrating between tenants, inventorying what you have, and confirming what migrates automatically versus what requires manual effort. This notebook guides you through discovery, scenario identification, and tool selection.

> **S2S Migration Journey — 3 Phases / 9 Steps**
>
> **Plan: 1. Discover** | 2. Strategize | 3. Design
>
> **Upgrade:** 4. Prepare | 5. Execute | 6. Integrate
>
> **Run:** 7. Expand | 8. Enable | 9. Optimize

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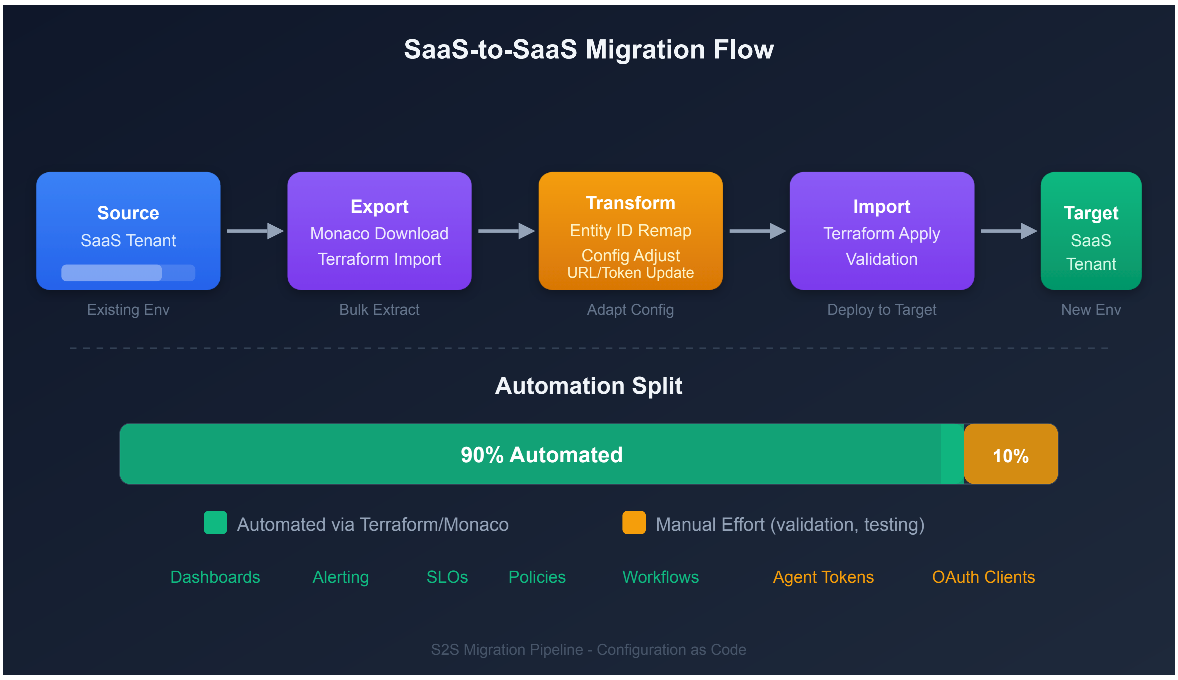
Prerequisites

Requirement	Details
Source Tenant	Active Dynatrace SaaS environment with administrator access
Target Tenant	Provisioned Dynatrace SaaS environment (or planning to provision)
API Tokens	Tokens with `entities.read`, `settings.read` scopes on source tenant
CLI Tools	Monaco CLI v2.x, Terraform v1.5+ (for IAM only)
Stakeholder Access	Ability to document and share findings with decision-makers



1. Why Migrate Between SaaS Tenants

Migrating between Dynatrace SaaS tenants is fundamentally different from migrating from Managed to SaaS. Both source and target are Gen3 Grail-powered environments, which simplifies some aspects (no architecture upgrade) but introduces unique challenges (entity ID remapping, historical data gaps, parallel operation).

Migration Scenarios

Scenario	Description	Example
Tenant Consolidation	Multiple SaaS tenants merged into a single tenant	Merging 5 regional tenants into 1 global tenant after M&A
Account Restructuring	Redistribute environments across different account structures	Spinning off a business unit into its own Dynatrace account
Regional Relocation	Move to a different SaaS region for compliance or performance	Moving from US-hosted to EU-hosted SaaS cluster for GDPR
License Restructuring	Restructure DPS allocation across tenants	Moving from multiple small tenants to a single enterprise agreement
Cloud Transformation	AWS → Azure, Azure → AWS, or cross-cloud consolidation	Rebuilding workloads on a new cloud provider with new monitoring
Environment Promotion	Promote a staging or POC tenant to production	Converting a successful POC into the production monitoring tenant

> **Key Difference from M2S:** In a Managed-to-SaaS migration, the SaaS Upgrade Assistant handles most of the heavy lifting. For SaaS-to-SaaS, there is **no automated assistant** — you use Monaco, Terraform, and the Settings API to export and reimport configuration.

S2S-Specific Order of Operations

The S2S migration follows an 11-step order of operations within the 9-step framework:

Step	Action	Phase
1	**Assess** — Inventory source tenant	Plan
2	**Provision** — Target SaaS tenant and access (SSO/IAM)	Plan
3	**Export** — Configuration from source tenant	Upgrade

Step	Action	Phase
4	**Migrate** — Configuration to target tenant	Upgrade
5	**Rebuild** — Dashboards, SLOs, alerts	Upgrade
6	**Redirect** — OneAgents and operators to target	Upgrade
7	**Reconnect** — Cloud integrations and extensions	Upgrade
8	**Migrate** — Any remaining configuration	Upgrade
9	**Validate** — Data flow and performance	Run
10	**Cutover** — Full switch to target tenant	Run
11	**Decommission** — Source tenant	Run

2. What Migrates and What Does Not

Understanding portability constraints upfront prevents surprises during execution.

Configuration That CAN Be Migrated (with tooling)

Configuration Type	Tool	Notes
Gen3 settings (Settings 2.0)	Monaco, Terraform	Bulk export/import via `monaco download`
Dashboards and notebooks	Monaco (documents type), Terraform	Entity ID references must be updated in target
Workflows and automations	Monaco (automations type), Terraform	Requires OAuth client credentials
OpenPipeline rules	Monaco (openpipeline type), Terraform	Bucket references must match target
Grail bucket configuration	Monaco (bucket type), Terraform	Retention policies may differ
SLO definitions	Monaco (settings/slo-v2 type), Terraform	Metric expressions may reference entity IDs

Configuration Type	Tool	Notes
Management zones	Monaco, Terraform	Zone rules and entity assignments
Auto-tagging rules	Monaco, Terraform	Tag rules and conditions
Service detection rules	Monaco, Terraform	Custom service naming and merging
Request attributes	Monaco, Terraform	Capture rules for request metadata
Synthetic monitors	Monaco, Terraform	Requires classic API token (v1.88.0+)
Segments	Monaco, Terraform	Download, deploy, and delete
Classic dashboards	Monaco, Terraform	JSON export/import; entity IDs must be updated

What CANNOT Be Migrated

Item	Reason	Impact
Historical metrics, logs, traces	Stored in source tenant's Grail	Run parallel tenants during transition
Entity IDs	Unique per tenant, auto-generated	Remap references in dashboards/SLOs
Dynatrace Intelligence baselines	Learned from source data	Requires 2–4 weeks to retrain
Session replay recordings	Bound to source tenant	Accept gap or extend parallel period
Problem history	Stored in source tenant	Export key problems as documentation
Credential Vault secrets	Security — secrets cannot be exported	Recreate in target Credentials Vault
API tokens	Security — environment-specific	Generate new tokens in target

Item	Reason	Impact
OAuth client secrets	Security — cannot be exported	Create new OAuth clients in target
Synthetic execution history	Bound to source tenant	Execution data starts fresh
Smartscape topology history	Computed per-tenant	Rebuilds automatically in target

> **Important:** Historic data does **not** migrate. Plan for a parallel-run period (typically 2–4 weeks) where both tenants receive data so the target tenant accumulates its own baselines and history.

3. Entity Inventory

Run these DQL queries against the **source** tenant to understand the full monitoring footprint. This inventory drives license sizing, identifies migration complexity, and serves as the validation baseline after cutover.

Host Inventory

```
// Host inventory by cloud provider and OS
// Note: cloud provider fields (awsNameTag, azureResourceGroupName,
gcpProjectId)
// are only present on hosts monitored in those cloud environments
fetch dt.entity.host
| fieldsAdd provider = if(isNotNull(awsNameTag), then: "AWS",
    else: if(isNotNull(azureResourceGroupName), then: "Azure", else: "On-
Premises"))
| summarize count = count(), by:{provider, osType}
| sort count desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST
// | fieldsAdd provider = if(isNotNull(awsNameTag), then: "AWS",
// else: if(isNotNull(azureResourceGroupName), then: "Azure", else: "On-
Premises"))
// | summarize count = count(), by:{provider, osType}
// | sort count desc
```

Kubernetes Cluster Inventory

```
// Kubernetes cluster inventory
fetch dt.entity.kubernetes_cluster
| fields entity.name, id
| sort entity.name asc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes K8S_CLUSTER
// | fields name, id
// | sort name asc
```

Service Inventory

```
// Service inventory by technology
fetch dt.entity.service
| summarize count = count(), by:{serviceType}
| sort count desc

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes SERVICE
// | summarize count = count(), by:{serviceType}
// | sort count desc
```

Application and Synthetic Inventory

```
// Application and synthetic monitor counts
fetch dt.entity.application
| summarize app_count = count()
| append [fetch dt.entity.synthetic_test | summarize synthetic_count =
count()]
```

ActiveGate Inventory

```
// ActiveGate inventory — query hosts with isActiveGate property
// Note: ActiveGate entity type name varies by deployment (active_gate or
environment_active_gate)
fetch dt.entity.host
| fieldsAdd entity.name
| summarize activeGateCount = count()
```

Configuration Change Audit

Understanding what has been actively changed in the last 30 days helps prioritize which configurations are actively managed versus stale.

```
// Audit log: recent configuration changes (last 30 days)
fetch logs, from:-30d
| filter matchesPhrase(log.source, "audit")
| filter contains(content, "settings")
| parse content, "JSON:json"
| fieldsAdd schemaId = json["schemaId"]
| summarize changes = count(), by:{schemaId}
| sort changes desc
| limit 20
```

Entity Summary Table

Record your findings in this table:

Entity Type	Count	Notes
Hosts	___	Include OS and cloud provider breakdown
Kubernetes Clusters	___	Cluster names and versions
Services	___	Include technology breakdown
Applications	___	Web and mobile
Synthetic Tests	___	HTTP, browser, scripted
ActiveGates	___	Environment AGs and cluster AGs
Process Groups	___	Total monitored process groups

4. Configuration Inventory

Beyond entities, you need a count of configuration objects to estimate migration effort. Use the tables below as a checklist.

Gen2 (Classic) Configuration

Category	API	Your Count	Typical Range
Management zones	Config API v1: `/managementZones`	___	5–30

Category	API	Your Count	Typical Range
Auto-tags	Config API v1: `/autoTags` or Settings 2.0	___	10–50
Alerting profiles	Config API v1: `/alertingProfiles`	___	5–20
Notification rules	Config API v1: `/notifications` or Settings 2.0	___	5–50
Calculated service metrics	Config API v1: `/calculatedMetrics/service`	___	10–50
Request attributes	Config API v1: `/requestAttributes` or Settings 2.0	___	5–30
Custom services	Config API v1: `/service/customServices`	___	5–20
Application detection rules	Config API v1 or Settings 2.0	___	5–20
Conditional naming rules	Config API v1 or Settings 2.0	___	5–15
Maintenance windows	Config API v1 or Settings 2.0	___	3–10
Classic dashboards	Dashboard API v1	___	20–200
Credential vault entries	Config API v1	___	5–20

Gen3 (Grail) Configuration

Category	API	Your Count	Typical Range
Settings 2.0 schemas (total)	Settings API	___	50–500
Dashboards (modern)	Document API	___	20–200

Category	API	Your Count	Typical Range
Notebooks	Document API	___	5–50
SLO definitions	SLO API / slo-v2	___	10–100
Synthetic monitors	Synthetic API	___	10–100
Workflows	Automation API	___	5–30
OpenPipeline rules	OpenPipeline API	___	1–10
Grail buckets	Bucket API	___	1–10
Segments	Segment API	___	1–20
K8s enrichment rules	Settings API: `builtin:kubernetes.metadata.enrichment`	___	1–20

```
// detected problem inventory — run on source tenant before migration
// High active problem counts indicate noise that will carry to the target
fetch dt.davis.problems, from:-30d
| summarize
    total = count(),
    active = countIf(event.status == "ACTIVE"),
    frequent = countIf(dt.davis.is_frequent_event == true),
    duplicate = countIf(dt.davis.is_duplicate == true)
| fieldsAdd triage_recommendation = if(active &gt; 500,
    then: "TRIAGE BEFORE MIGRATION — suppress frequent/duplicate events",
    else: "Manageable — review active problems during parallel period")
```

5. Detected Problem Triage and Configuration Debt

Discovery is not just about counting what you have — it is also about identifying what you should **not** migrate. Active detected problems and stale configuration carry over to the target tenant if not triaged, creating noise that obscures real issues during the critical parallel operation period.

Detected Problem Inventory

Run these queries against the source tenant to understand the current problem landscape. Large numbers of active problems (especially frequent/duplicate events) indicate configuration debt that should be resolved *before* migration — not after.

Signal	Action
Active problems > 500	Triage before migration — most are likely noise or duplicates
Frequent events > 50% of total	Suppress or tune anomaly detection before export
Problems older than 30 days	Investigate — likely stale or auto-resolved but not closed

Configuration Debt Cleanup Opportunity

Migration is the best time to leave legacy behind. Stale configuration inflates the export, slows Monaco deploy, and creates confusion in the target tenant.

Candidate	How to Identify	Recommendation
Disabled notification rules	`settings.read` audit — any notification with `enabled: false`	Do not migrate
Inactive synthetic monitors	Monitors with no executions in 30+ days	Triage — migrate only active monitors
Stale maintenance windows	Windows with past end dates or no recurrence	Do not migrate
Unused management zones	Zones with zero entity matches	Do not migrate
Legacy auto-tag rules	Tags that duplicate OneAgent attribute enrichment	Replace with primary tags at agent level

> **Lesson from real migrations:** One engagement discovered 366 maintenance windows in the source tenant — all with expired dates. Migrating them would have

> cluttered the target tenant with useless configuration. Triaging before export saved significant cleanup effort later.

5. Migration Tools Comparison

Tool	Best For	Strengths	Limitations
Monaco	Bulk config export/import: settings, documents, automations, buckets, segments, slo-v2, openpipeline, classic API	`monaco download` bulk export, `monaco deploy` with automatic dependency resolution, no HCL knowledge needed	No state management, no drift detection, cannot manage IAM
Terraform	IAM (policies, groups, bindings), ongoing infrastructure-as-code management	State tracking, drift detection via `terraform plan`, cross-platform resource references	Requires HCL knowledge, no bulk export equivalent
Settings API	Targeted, surgical changes to specific settings	Fine-grained programmatic control	Custom scripting required for large-scale migration

> **Note:** The SaaS Upgrade Assistant is for Managed-to-SaaS migrations only. It does **not** support SaaS-to-SaaS.

When You Need Terraform

Terraform is required **only** when you need to manage:

- **IAM policies, groups, and bindings** — Monaco cannot manage account-level IAM
- **State management and drift detection** — Monaco deploys are stateless

For everything else, Monaco and Terraform are functionally equivalent. Choose based on your team's existing expertise.

Recommended Approach

Use **Monaco for bulk configuration** and **Terraform for IAM only**:

```
# Step 1: Monaco download from source tenant
monaco download manifest.yaml --environment source-tenant

# Step 2: Update manifest to point at target tenant
# Step 3: Validate and deploy
monaco deploy manifest.yaml --environment target-tenant --dry-run
monaco deploy manifest.yaml --environment target-tenant

# Step 4: Use Terraform only for IAM
terraform apply -target=dynatrace_iam_policy.example
```

> **Download limitations:** Cloud provider credentials (AWS, Azure, K8s) can be deployed by Monaco but **not exported** via `monaco download`. These must be recreated manually in the target tenant regardless of tool choice.

6. The 90/10 Rule

The 90/10 rule is the defining reality of SaaS-to-SaaS migration:

Phase	Effort	What It Covers
Automated Export/Import (90% of config)	~10% of total effort	Settings 2.0, dashboards, SLOs, notification rules, enrichment rules, OpenPipeline
Manual Remediation (10% of config)	~90% of total effort	Entity ID remapping, webhook URL updates, IAM redesign, cloud integration reconfiguration, parallel validation

Why Manual Effort Dominates

- **Entity IDs change** between tenants — every dashboard filter, SLO metric expression, and notification rule that references an entity ID must be updated
- **Integrations are tenant-specific** — webhook URLs, cloud provider connections, and SSO configurations must be reconfigured

- **Historical data cannot move** — parallel operation is required to maintain continuity
- **Dynatrace Intelligence must relearn** — baselines take 2–4 weeks to stabilize in the target tenant

Items That Require Manual Attention

Item	Why It Cannot Be Automated
Entity ID references in dashboards	IDs are auto-generated per tenant
Entity ID references in SLO expressions	Metric selectors embed entity IDs
Webhook notification URLs	Network paths differ between tenants
Cloud provider credentials	Secrets cannot be exported
SSO/SAML configuration	Identity provider settings are tenant-specific
Synthetic private locations	ActiveGate-bound, environment-specific
Extensions 2.0	Neither Monaco nor Terraform supports export — reinstall from Hub

8. Step Completion Checklist

Before proceeding to **Step 2 — Strategize**, confirm that you have completed each item:

Checkpoint	Status
Migration scenario identified (consolidation, split, regional relocation, cloud transformation)	[ ]
Entity inventory complete (hosts, services, K8s clusters, applications, synthetics, ActiveGates)	[ ]
Configuration inventory complete (Gen2 counts + Gen3 counts)	[ ]
detected problem triage complete — frequent/duplicate events suppressed or tuned	[ ]

Checkpoint	Status
Configuration debt identified — stale maintenance windows, disabled rules, inactive monitors cataloged	[ ]
Non-portable items identified (credentials, tokens, entity IDs, historical data)	[ ]
Migration tools selected (Monaco for bulk config + Terraform for IAM)	[ ]
90/10 manual items documented	[ ]
Stakeholders informed of historical data limitation and parallel-run requirement	[ ]

Next Step

> **S2S-02: Step 2 — Strategize** — Define your migration approach, timeline, and risk assessment. Choose between big-bang and phased migration, plan your parallel-run period, and identify dependencies and risks.

Additional Resources

- Monaco Configuration as Code
- Dynatrace Terraform Provider
- Settings API
- Grail Data Lakehouse
- DQL Reference

Summary

In Step 1, you:

- Identified your migration scenario (consolidation, split, regional relocation, cloud transformation, or environment promotion)
- Documented what configuration migrates with tooling versus what requires manual recreation

- Completed an entity inventory (hosts, services, K8s clusters, applications, synthetics, ActiveGates)
- Completed a configuration inventory (Gen2 classic + Gen3 Grail counts)
- Selected migration tools (Monaco for bulk config, Terraform for IAM)
- Understood the 90/10 rule: 90% of config migrates automatically, but the remaining 10% takes 90% of the effort

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